



MEASURING AI ABILITY TO COMPLETE LONG SOFTWARE TASKS

Time Horizon Doubling Every 7 Months

Source: Model Evaluation & Threat Research (METR), March 2025



A New Metric: 50% Task-Completion Time Horizon

The time-horizon metric quantifies AI capability by measuring the duration of tasks (in human time) that an AI model can complete with a **50% success rate**.

- **Step 1:** Assemble a diverse suite of 170 software & research tasks.
- **Step 2:** Time skilled human professionals and measure AI agent success rates.
- **Step 3:** Fit a logistic model to estimate the human-time duration where AI succeeds half the time.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

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170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Suite combines HCAST (97 tasks), RE-Bench (7 tasks), and SWAA (66 tasks), backed by >2,500 hours of human baselines.

EXAMPLE TASK	HUMAN TIME	DESCRIPTION
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cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1h 50m (o3)

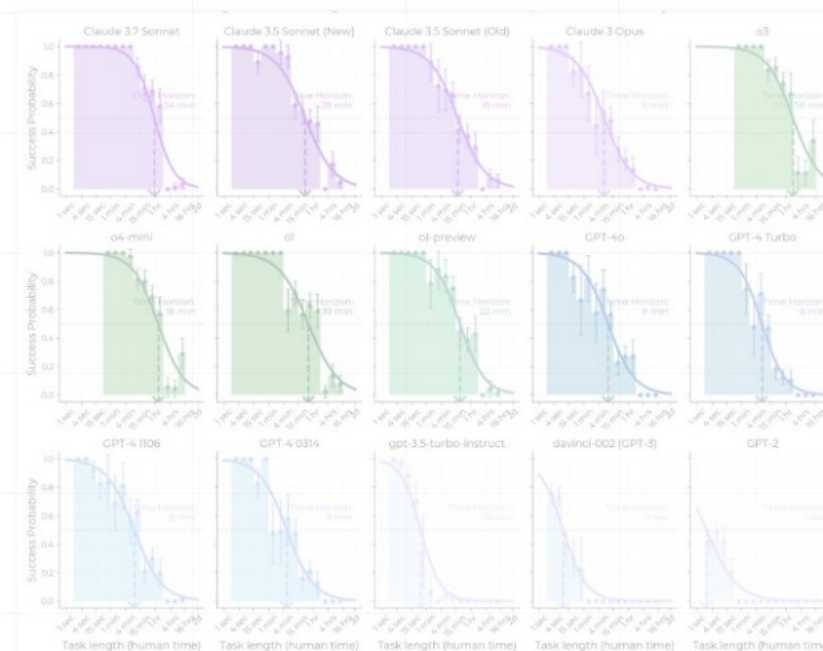
Exponential Growth (2019-2025)

- **Doubling Time:** ~207 days (~7 months)
- **High Predictability:** $R^2 = 0.97$

GPT-2 (2019): ~2 seconds

GPT-4 (2023): ~5 minutes

o3 (2025): ~1 hour 50 minutes



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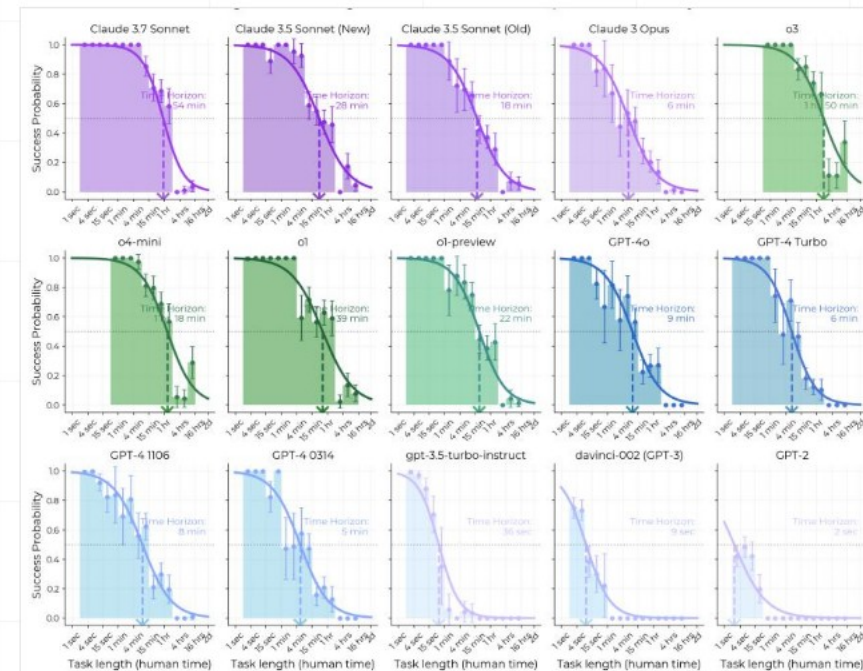
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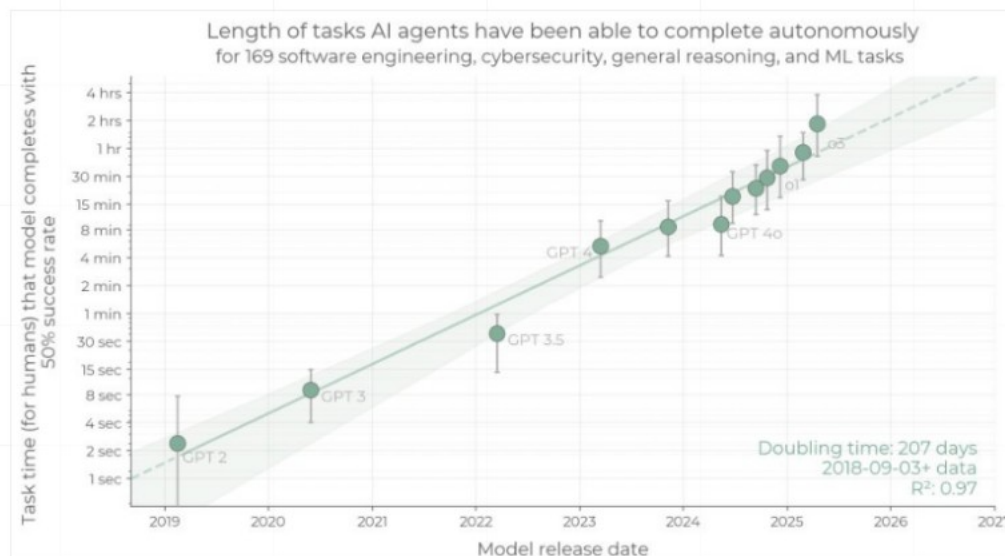
o3 (2025): ~1 hour 50 minutes



Success Probability Drops Sharply as Task Length Increases

While overall capability is growing, success rates follow a consistent curve across all models: rapidly degrading as tasks require more time and steps.

Frontier models like **Claude 3.7 Sonnet (54m)** and **o3 (1h50m)** push the success curve significantly rightward compared to older generations, sustaining reliability over much longer horizons.



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